EPS Payment API

Reliable Money Transfer

Disclaimer: This document contains confidential information about EPS Payment API, which is provided for the purpose of permitting the recipient to evaluate the proposal submitted herewith. In consideration of the recipient of this document, the recipient agrees to maintain such information in confidence and to not reproduce or otherwise disclose this information to any person outside of the recipient's employees directly responsible for the evaluation of its contents. There is no obligation to maintain the confidentiality of any information from EPS Payment API or becomes publicly known through no fault of the recipient, or is received without obligation of confidentiality from a third party owing no obligation of confidentiality to EPS Payment API.

2011

Document Revision History

Date	Interface Version	Document Version	Description	Author
12/12/2011	1.0	1.0	Newly written document	Muhammad Haseeb Bhatti
15/12/2011	1.0	1.1	• AgentCode, IdNumber & PayBy information added in GetPaymentDetails response.	Waqas Tariq Dar
22-05-2012	1.0	1.1.1	• Explanation added for Sample Response Codes.	Waqas Tariq Dar
13-06-2012	1.0	1.1.2	• *PayBy* attribute in *GetPaymentDetails* is replaced by *PayDescription* attribute. • *PayBy* attribute is removed from the REQUEST of PayMoney operation. • *ReceiverName* attribute added in the REQUEST of *PayMoney*. Response Code(211) also added for this attribute. • Size of *ReceiverName* in *GetPaymentDetails* & *PayMoney* is 150 characters now. • Size of *BranchCode* attribute in *GetBranchNetwork* is 20 characters now.	Waqas Tariq Dar

Table of Contents

1. Electronic Payment System (EPS) 4
2. EPS Payment API 4
3. Connectivity 4
4. Security 4
5. Flow of a Transaction 4
6. API URL 6
7. API Operations 6
   7.1 Get Payment Details 6
   7.2 Pay Money 7
   7.3 Get Branch Network 9
   7.4 Change Password 10
8. Sample Response Codes 11
9. Contact Details 11

EPS Payment API

1. Electronic Payment System (EPS)

Electronic Payment System is a fast, reliable and secure system to transfer money worldwide. EPS is a complete web-based, end-to-end, multi-currency and multi-country money transfer software system.

2. EPS Payment API

This document provides an overview of Electronic Payment System's Payment API. EPS Payment API provides features for payment detail check, pay money against a particular payment, get branch network and change password. The system is exposed via a web service that operates in real time.

3. Connectivity

The service is made available to the Payee Party systems via a Web Service. To access this web service the Payee Party is provided a user name and a password which is required to be supplied for each transaction request. Because of the fact that web services are platform independent, the system allows for integration with heterogeneous systems developed in any technology.
It has a SOAP based interface which is a worldwide standard.

4. Security

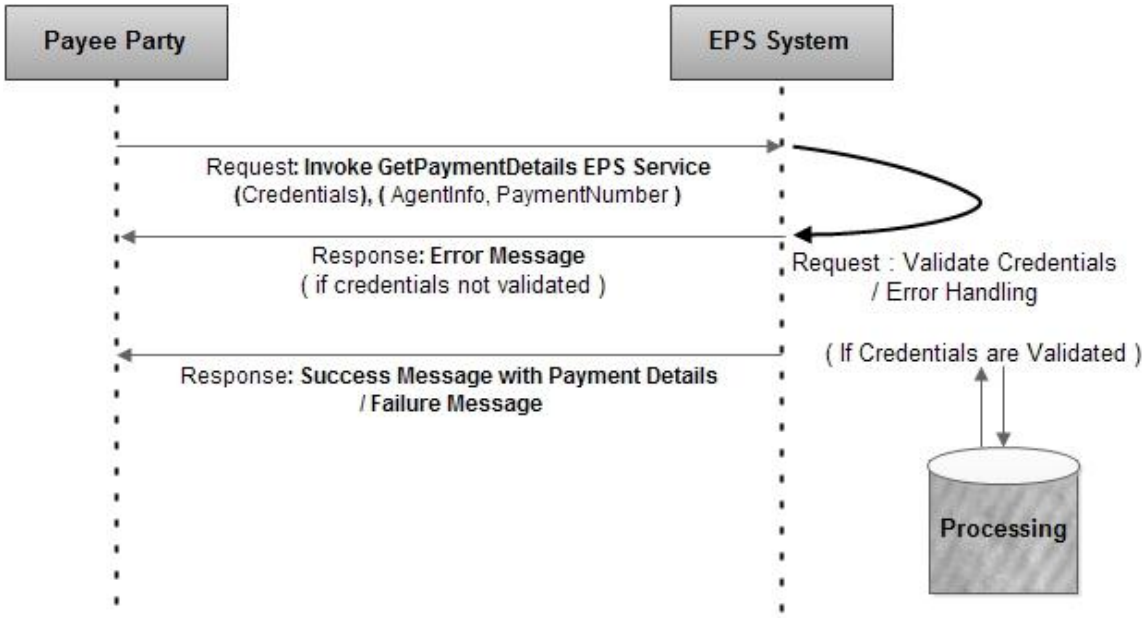
Payee Party is provided a user name (Agent Code) and a password which is mandatory for every transaction and transactions are carried out only after the Payee Agent is authenticated using the provided Agent Code and Password. Following are the salient security features of the system.

- The system employs industry standard security techniques to prevent disclosure of sensitive data and to avoid malicious attacks.
- The system is capable of using public-private key encryption between the client and the server.
- The public / private key pairs are changed periodically.
- All user passwords are encrypted using one way encryption.
- All communication is done over Secure Socket Layer (SSL).
- Values input into every field are validated before use.
- The data access layer will be responsible for SQL injection attacks(i.e., hackers attempting to enter SQL statements through application UI fields) (By Using Hibernate)

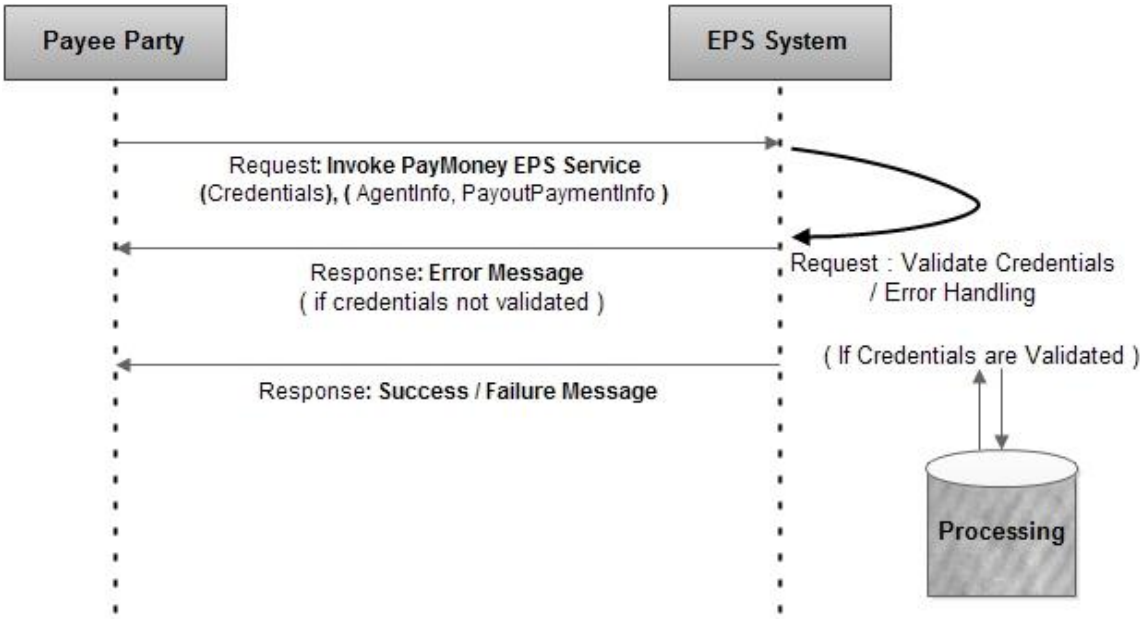
5. Flow of a Transaction

Given below is the flow of EPS Payment Web Service initiated by the Payee Party.

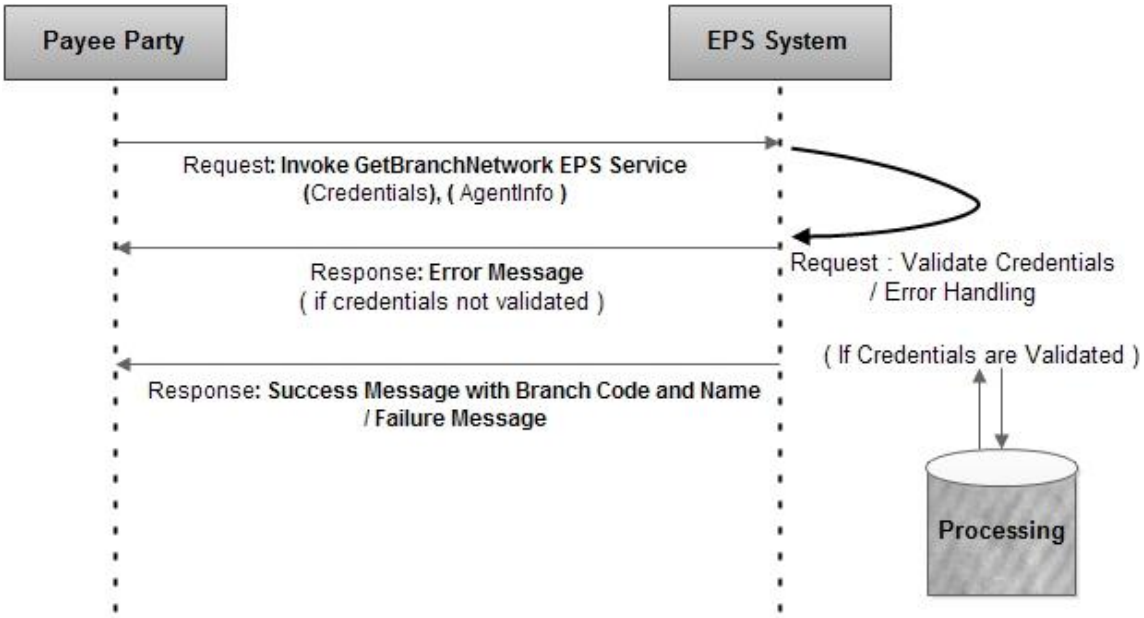
Process Flow – Get Payment Details



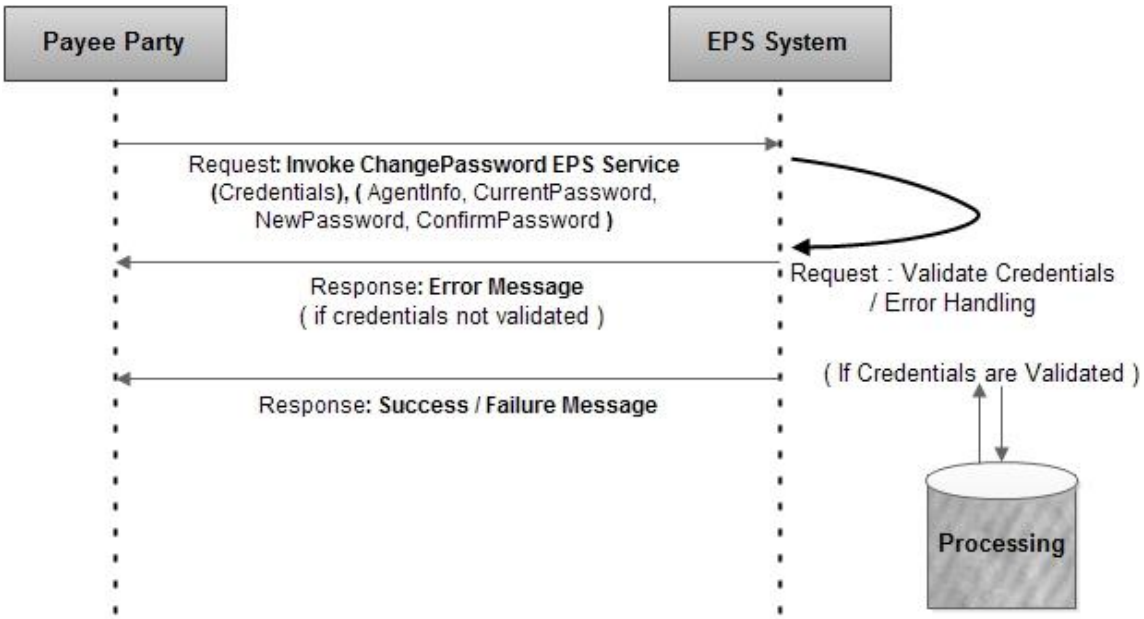
Process Flow – Pay Money



Process Flow – Get Branch Network



Process Flow – Change Password



6. API URL

- **http://server2.dex-intl.com/branches/eps/pay/service.php?wsdl**

7. API Operations

Following is the detailed description of all operations implemented in the API along with the length and types of their parameters. With the help of these operations payee party can easily get the payment details, pay money, get branch network, and change their password.

7.1 Get Payment Details

The use of this operation is to get the detail of a particular payment. This will help the payee party to see the payment details which they want to pay. A success/failure message along with Response Code and some other payment credentials will be sent to the payee party as a response of this operation.

7.1.1 Sample Input

- **GetPaymentDetails → Request from Payee Party to EPS**

```
<SOAP-ENV:Body>
<ns1:GetPaymentDetails>
<AgentInfo xsi:type="ns2:AgentInfo">
<AgentCode xsi:type="xsd:string">PK003</AgentCode>
<Password xsi:type="xsd:string">1</Password>
</AgentInfo>
<PaymentNumber xsi:type="xsd:string">0307101050086</PaymentNumber>
</ns1:GetPaymentDetails>
</SOAP-ENV:Body>
```

	Field Name	Size	Type	Description	O/M
Agent Info	AgentCode	20	String	Agent Code given to Payee Party	M
	Password	20	String	Password of Agent given to Payee Party	M
	PaymentNumber	20	String	Unique Payment Number	M

7.1.2 Sample Output

- **GetPaymentDetails ← Response from the EPS to the Payee Party**

```
<SOAP-ENV:Body>
<ResponseCode xsi:type="xsd:numeric">208</ResponseCode>
<Message xsi:type="xsd:string">Payment Found</Message>
<ns1:PaymentDetail xmlns:ns1="urn:GetPaymentDetails">
<PaymentNumber xsi:type="xsd:string">0307101050086</PaymentNumber >
<Status xsi:type="xsd:string">Payment Already Paid</Status >
<AgentCode xsi:type="xsd:string">PK003</AgentCode>
<IdNumber xsi:type="xsd:string">123</IdNumber>
<PayBy xsi:type="xsd:string">0001</PayBy>
<TransactionDate xsi:type="xsd:datetime">2011-06-06</TransactionDate >
<Amount xsi:type="xsd:string">20290</Amount>
</ns1: PaymentDetail >
```

<ns1:SenderDetail xmlns:ns1="**urn:GetPaymentDetails**">
<SenderName xsi:type="**xsd:string**">**Mohsin**</ SenderName >
<SenderPhone xsi:type="**xsd:numeric**">**03457895854**</ SenderPhone >
<SenderAddress xsi:type="**xsd:string**">**Model Town**</ SenderAddress >
<SenderCountry xsi:type="**xsd:string**">**Pakistan**</ SenderCountry >
</ns1: SenderDetail >
<ns1: ReceiverDetail xmlns:ns1="**urn:GetPaymentDetails**">
<ReceiverName xsi:type="**xsd:string**">**Haseeb**</ ReceiverName >
<ReceiverAddress xsi:type="**xsd:string**">**Wapda Town**</ ReceiverAddress >
<ReceiverPhone xsi:type="**xsd:numeric**">**03456250585**</ ReceiverPhone >
<ReceiverCNIC xsi:type="**xsd:numeric**">**3410139904569**</ ReceiverCNIC >
</ns1: ReceiverDetail >
</SOAP-ENV:Body>

	Field Name	**Size**	**Type**	**Description**	**O/M**
	ResponseCode	3	Numeric	Response Code explaining result of request	M
	Message	100	String	Text to describe the response of the request	M
Payment Detail	PaymentNumber	20	String	Unique payment number	M
	Status	100	String	Current Status of the payment.	M
	AgentCode	20	String	Agent Code of Payee Party	M
	IdNumber	30	String	Id Number of the Payee	M
	PayDescription	30	String	Payee Branch Code.	M
	Transaction Date		DateTime	Date of the Transaction	M
	Amount	15	Numeric	Amount payable by the payee party	M
Sender Detail	SenderName	50	String	Name of the Sender	M
	SenderPhone	30	String	Phone Number of the Sender	M
	SenderAddress	40	String	Address of the Sender	M
	SenderCountry	30	String	Country Name of the Sender	M
Receiver Detail	ReceiverName	150	String	Name of the Receiver	M
	ReceiverAddress	50	String	Address of the Receiver	M
	ReceiverPhone	30	String	Phone Number of the Receiver	M
	ReceiverCNIC	30	String	CNIC Number of the Receiver	M

7.2 Pay Money

Payee party will use this operation to pay the payment of EPS. A success/failure message along with the Response Code and Payment Number will be sent to the payee party as a response of this operation.

7.2.1 Sample Input

- **PayMoney → Request from Payee Party to EPS**

```
<SOAP-ENV:Body>
<ns1:PayMoney>
<AgentInfo xsi:type="ns2:AgentInfo">
<AgentCode xsi:type="xsd:string">PK003</AgentCode>
<Password xsi:type="xsd:string">1</Password>
</AgentInfo>
<PayoutPaymentInfo xsi:type="ns2:PayoutPaymentInfo">
<PaymentNumber xsi:type="xsd:string">0307101050086</PaymentNumber>
<PayDescription xsi:type="xsd:string">bank</PayDescription>
<IdNumber xsi:type="xsd:string">123</IdNumber>
<PayBy xsi:type="xsd:string">0001</PayBy>
</PayoutPaymentInfo>
</ns1:PayMoney>
</SOAP-ENV:Body>
```

	Field Name	Size	Type	Description	O/M
Agent Info	AgentCode	20	String	Agent Code given to Payee Party	M
	Password	20	String	Password of Agent given to Payee Party	M
Payout Payment	PaymentNumber	20	String	Unique Payment Number	M
	PayDescription	30	String	Payee Branch Code.	M
	IdNumber	30	String	Id Number of the Payee	M
	ReceiverName	150	String	Name of the Receiver	M

7.2.2 Sample Output

- **PayMoney ← Response from EPS to Payee Party**

```
<SOAP-ENV:Body>
<ns1:PayMoneyResponse xmlns:ns1="urn:PayMoney">
<ResponseCode xsi:type="xsd:numeric">104</ResponseCode>
<Message xsi:type="xsd:string">Error: System Can't pay this payment, please call
GetPaymentDetails for more details</Message>
<PaymentNumber xsi:type="xsd:string">0307101050086</PaymentNumber>
</ns1:PayMoneyResponse>
</SOAP-ENV:Body>
```

	Field Name	Size	Type	Description	O/M
Pay Money Response	ResponseCode	3	Numeric	Response Code explaining result of request	M
	Message	100	String	Text to describe the response of the	M

				request	
	PaymentNumber	20	String	Unique Payment Number	M

7.3 Get Branch Network

This operation is used to see a list of branches of a Payee Agent.

7.3.1 Sample Input

- **GetBranchNetwork → Request from Payee Party to EPS**

```
<SOAP-ENV:Body>
<ns1:GetBranchNetwork>
<AgentInfo xsi:type="ns2:AgentInfo">
<AgentCode xsi:type="xsd:string">PK003</AgentCode>
<Password xsi:type="xsd:string">1</Password>
</AgentInfo>
</ns1:GetBranchNetwork>
</SOAP-ENV:Body>
```

Agent Info	Field Name	Size	Type	Description	O/M
	AgentCode	20	String	Agent Code given to Payee Party	M
	Password	20	String	Password of Agent given to Payee Party	M

7.3.2 Sample Output

- **GetBranchNetwork ← Response from EPS to Payee Party**

```
<SOAP-ENV:Body>
<ns1:Array xmlns:ns1="urn:Array">
<BranchCode xsi:type="xsd:string">104</BranchCode>
<BranchName xsi:type="xsd:string">Bank Square Branch</BranchName>
</ns1:Array>
</SOAP-ENV:Body>
```

Array	Field Name	Size	Type	Description	O/M
	BranchCode	20	String	Branch Code of the Payee Party	M
	BranchName	30	String	Branch Name of the Payee Party	M

7.4 Change Password

This operation is used to change the password of the Payee Party. A success or failure message will be sent to the payee party in response of this operation.

7.4.1 Sample Input

- **ChangePassword → Request from Payee Party to EPS**

```
<SOAP-ENV:Body>
<ns1: ChangePassword>
<AgentInfo xsi:type="ns2:AgentInfo">
<AgentCode xsi:type="xsd:string">PK003</AgentCode>
<Password xsi:type="xsd:string">1</Password>
</AgentInfo>
<CurrentPassword xsi:type="xsd:string">123</CurrentPassword>
<NewPassword xsi:type="xsd:string">456</NewPassword>
<ConfirmPassword xsi:type="xsd:string">456</ConfirmPassword>
</ns1: ChangePassword>
</SOAP-ENV:Body>
```

	Field Name	Size	Type	Description	O/M
Agent Info	AgentCode	20	String	Agent Code given to Payee Party	M
	Password	20	String	Password of Agent given to Payee Party	M
ChangePassword Info	CurrentPassword	20	String	Password of the Payee Party to interact with the EPS Payment System.	M
	NewPassword	20	String	New Password to replace with the old Password	M
	ConfirmPassword	20	String	Re-Type the New Password	M

7.4.2 Sample Output

- **ChangePassword ← Response from EPS to Payee Party**

```
<SOAP-ENV:Body>
<ResponseCode xsi:type="xsd:numeric">210</ResponseCode>
<Message xsi:type="xsd:string">Password Changed Successfully </Message>
</SOAP-ENV:Body>
```

	Field Name	Size	Type	Description	O/M
	ResponseCode	3	Numeric	Response Code explaining result of request	M
	Message	100	String	Text to describe the response of the request	M

8. Sample Response Codes

Response Code	Message	Explanation
100	Query Successful.	Request processed successfully.
102	Payment Number Not Found.	Payment not found in our system.
103	Client Input Error*	Any error in the web service request message will be considered a *Client Input Error*, i-e "Please provide Agent Password for login" and "PaymentNumber is INVALID. Please provide VALID PaymentNumber to view its Status" are examples of *Client Input Error*.
104	Server Side Error*	Any error/restriction in our system which can stop the processing of your request is considered as Server Side Error, like "Error while running query in database".
200	Payment is paid successfully	Payment is marked as *Paid* in our system.
206	Not ready for payment. Please contact to Customer Service Centre.	It indicates that there is some problem with the Payment which you are going to pay.
207	Ready to be Paid.	This message means that everything seems OK, you can make a Payment request to our web service with the desired Payment Number.
208	Payment is Paid Already.	Payment is already marked as *Paid* in our system.
209	Kindly Change Your Password.	Indication to change your password.
210	Password Changed Successfully.	Your password changed successfully in our system.
211	ReceiverName mismatched.	If the ReceiverName has been changed in our system.

Note: * sign means that Description might be changed upon the type of error.

Contact Details:
SoftShack (Pvt) Ltd.
http://softshack.com.pk

Software Project Manager
Hafiz Tariq Aziz
Email: hafiztariq@gmail.com
Phone: +92-302-8743535